	(27)	0.7	**2.0**
	(28)	0.7	
	(29)	0.2	
합계			**10,0**

문제 3 (10점)
수소 원자의 보어 모형

1. .

R :

$$m_e\left(\frac{2\pi}{T}\right)^2 R = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2}, \qquad (1)$$

T .

:

$$\frac{a^3}{T^2} = \frac{1}{16\pi^3\varepsilon_0}\frac{e^2}{m_e}, \qquad (2)$$

a .

.

$a = r_0/2$. :

$$t_1 = \frac{T}{2} = \sqrt{\frac{\pi^3 m_e \varepsilon_0 r_0^3}{2e^2}} = 2.46\times10^{-17}\ \text{c}. \qquad (3)$$

2.

:

$$m_e\frac{\mathrm{v}^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}, \qquad (4)$$

:

$$\mathrm{v} = \sqrt{\frac{1}{4\pi\varepsilon_0}\frac{e^2}{m_e r}}, \qquad (5)$$

3. (5) :

$$L = m_e \mathrm{v}\, r = \sqrt{\frac{m_e r e^2}{4\pi\varepsilon_0}}. \qquad (6)$$

4.

. $L = n\hbar$. (6) :

$$r_n = \frac{4\pi\varepsilon_0 n^2\hbar^2}{m_e e^2}. \qquad (7)$$

5. (7) $n=1$.

$$r_1 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} = 5.19\times10^{-11}\ \text{м}. \qquad (8)$$

6. .

(5) :

$$E = \frac{m_e v^2}{2} - \frac{e^2}{4\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r}. \qquad (9)$$

(7) :